

Harsha Talapaka

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EDUCATION

The George Washington University, School of Engineering & Applied Science
Master of Science in Computer Science (Machine Intelligence)

Washington, DC
May 2025

- **Coursework:** Algorithms, Data Mining, Machine Learning, Neural Networks, Statistical Analysis, Cloud Computing, Trustworthy AI.

Symbiosis International University
Bachelor of Technology in Electronics and Telecommunication
• *Minor in Artificial Intelligence & Machine Learning.*

Pune, MH, India
June 2023

TECHNICAL SKILLS

- **Languages:** Python (Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, HuggingFace), SQL (Advanced, PostgreSQL, BigQuery), TypeScript, Bash/Linux.
- **AI, ML & Data:** LangChain, LangGraph, RAG, Agentic Workflows, LLM Evaluation, LangSmith, A/B Testing, Causal Inference, dbt, Airflow, FastAPI, MLflow.
- **Infra & Cloud:** AWS (S3, EC2, SageMaker), GCP (BigQuery, Cloud Run, Vertex AI), Docker, Kubernetes, PostgreSQL, Tableau, Power BI, Git, CI/CD.

RELEVANT WORK EXPERIENCE

Holiday Channel

Remote, CO, USA

Data Scientist

June 2025 - Present

- Owned the full lifecycle of a production AI system at Holiday Channel over a 6.5M+ item corpus end to end, from problem framing and architecture through model development, deployment, and iteration, building an XGBoost model achieving over 90 percent accuracy.
- Designed and built a production LLM and RAG pipeline using LangChain and LangGraph, owning retrieval, generation, and evaluation personally, and reducing end to end latency by 30 percent through systematic profiling.
- Built scalable ETL and data pipelines using Python, SQL, and dbt with automated quality checks at every stage, ensuring clean, trustworthy data for every downstream model, dashboard, and report.
- Designed A/B testing and causal inference frameworks to measure the real impact of product, model, and pipeline changes, delivering findings that accelerated roadmap execution by 25 percent.
- Prioritized work and made build versus defer decisions based on stakeholder needs and technical feasibility, translating ambiguous business requirements into structured, shippable technical scope.
- Communicated technical findings and product trade offs clearly to both technical and non technical stakeholders through Tableau and Power BI dashboards, translating complex system behavior into decisions people could act on.

Forage AI

Remote, NJ, USA

Intern, Technical QA Data

January 2023 - May 2023

- Built Python and Selenium based ETL pipelines to extract structured data from client-specified web sources, loading validated records into PostgreSQL for downstream analysis.
- Wrote SQL queries to cross-check extracted records against source data, catching field-level mismatches and formatting inconsistencies before they reached the client deliverable.
- Analyzed recurring data quality issues across ingestion batches, documenting root causes and proposing validation rule changes that reduced repeat errors in later extraction runs.

TECHNICAL PROJECTS

Eluno OMS | FastAPI | PostgreSQL | React | LightGBM | Groq API

May 2026

- Owned a full-stack AI-assisted order management product end to end: scoped the spec, designed the PostgreSQL schema, built the FastAPI backend, trained a LightGBM SLA-breach model, wired in Groq for LLM intake parsing, and shipped a React dashboard, deployed live on Render in 48 hours.
- GitHub: github.com/harshaa1231/Eluno_TakeHome_AI_Product_Engineer

Graph-Based NL-to-SQL Agent | FastAPI | Gemini API | NetworkX

January 2026 - February 2026

- Agentic NL-to-SQL system with multi-step reasoning, schema graph traversal, and a two-layer guardrail architecture, with all outputs verified against ground truth before submission.
- GitHub: github.com/harshaa1231/Dodge-AI---Hiring-Assessment

RAG Benchmark, Strategy A vs B vs Hybrid | FAISS | BM25 | Docker

April 2026

- Benchmarked three retrieval strategies with MRR@3, NDCG@3, and cosine drift as an early warning signal; good expansion lifted NDCG +54%, and drift over 0.55 reliably predicted retrieval failure.
- GitHub: github.com/harshaa1231/RAG-ASSESSMENT-TELEPORT

TECHNICAL WRITING

- Published 7 technical ML articles on Medium (<https://medium.com/@hv6700030>) covering LLM evaluation, RAG systems, NLP pipelines, GANs, sentiment classification, and model behavior analysis; each article includes reproducible code.